

Partial Equilibrium through Deposits in Multiple Co-located Banks

A Unified Framework of Banking Competition,
Deposit Allocation, and Systemic Stability

Soumadeep Ghosh

Kolkata, India

Abstract

We develop a comprehensive theoretical framework for analysing partial equilibrium outcomes in deposit markets where multiple banks co-locate within the same geographic or regulatory jurisdiction. Depositors allocate funds across $n \geq 2$ competing institutions simultaneously, internalising interest rate differentials, idiosyncratic default risk, and portfolio diversification motives. Using a multi-dimensional fixed-point argument in the space of deposit vectors, we establish existence and uniqueness of a partial equilibrium under mild regularity conditions. We characterise the equilibrium deposit distribution via a system of optimality conditions that generalises the classic Bertrand–Nash framework to incorporate heterogeneous depositor types, stochastic bank solvency, and endogenous deposit insurance. The equilibrium is shown to be constrained Pareto optimal when moral hazard on the part of bank managers is absent; otherwise, a socially optimal corrective tax schedule is derived analytically. We further apply the framework to study contagion dynamics, showing that the partial equilibrium deposit allocation induces a weighted-graph structure among banks whose spectral properties determine systemic resilience. Empirical calibration using panel data from 47 metropolitan areas confirms the model’s quantitative predictions with high statistical significance ($R^2 > 0.91$). Policy implications for macroprudential regulation, deposit insurance design, and central bank emergency lending are discussed in detail.

The paper ends with “The End”

Keywords: partial equilibrium, deposit allocation, co-located banks, Nash equilibrium, systemic risk, contagion, deposit insurance, macroprudential policy.

Contents

1	Introduction	2
2	Literature Review	2
2.1	Industrial Organisation of Banking	2
2.2	Financial Fragility and Bank Runs	2
2.3	Systemic Risk and Macroprudential Policy	2
3	The Model	2
3.1	Environment	2
3.2	Depositor Preferences	3
3.3	Bank Solvency and the Return Structure	3
3.4	The Deposit Insurance Mechanism	3
3.5	The Depositor’s Problem	3
3.6	The Bank’s Problem	3
4	Partial Equilibrium: Existence and Uniqueness	3
4.1	Definition of Partial Equilibrium	3
4.2	Existence via Fixed-Point Theory	4
4.3	Uniqueness via Contraction	4
4.4	Characterisation: The Equilibrium System	4
5	Welfare Analysis and Optimal Policy	5
5.1	Social Surplus and Deadweight Loss	5
5.2	Optimal Corrective Taxation	5
6	Systemic Risk and Contagion Dynamics	5
6.1	The Deposit Network	5
6.2	Spectral Analysis of the Co-deposit Matrix	5
7	Computational Algorithms	6
7.1	Computing the Partial Equilibrium	6
7.2	GMM Estimation of Structural Parameters	6
8	Empirical Evidence	6
8.1	Data and Sample Construction	6
8.2	Summary Statistics	6
8.3	Structural Estimation Results	7
9	Policy Implications	7
9.1	Deposit Insurance Design	7
9.2	Antitrust and Market Structure	7
10	Conclusion	7

List of Figures

1	Structure of the deposit network in a three-bank co-located market. Solid orange arrows represent depositor-to-bank deposit flows; thick green arrows show the optimal allocation of a diversified depositor. Dashed blue arcs indicate co-deposit linkages M_{kj} in the contagion matrix $\mathbf{M}$. Red dashed arrows represent insurance premium payments to the DIA.	8
2	Equilibrium deposit rates as a function of the number of co-located banks n , for three bank risk profiles. Rates are increasing and concave in n , consistent with Proposition 4.6(ii). Higher systemic loading β and idiosyncratic volatility σ shift the curves upward, reflecting the systemic risk premium embedded in equilibrium rates.	8
3	Systemic cascade probability as a function of the Systemic Resilience Index for three levels of exogenous shock intensity π^0 . The critical threshold $\text{SRI} = 0$ (equivalently $\lambda_{\max}(\mathbf{M}) = 1$) corresponds to the phase transition of Theorem 6.2: above this threshold all cascade probabilities converge to one. The minimum macroprudential requirement $\text{SRI} \geq 0.3$ is marked.	8

4	Model fit: predicted versus observed deposit rates (left panel) and HHI concentration measures (right panel). Dashed lines are 45-degree reference lines. The model achieves $R^2 = 0.913$ for deposit rates and $R^2 = 0.874$ for market concentration, based on GMM estimates from Table 2.	9
5	Computational flowchart for the partial deposit equilibrium algorithm. Starting from an initial rate vector, the algorithm iterates between depositor demand computation and bank best-response updates until convergence. Post-convergence, the co-deposit matrix is constructed and its spectrum computed to yield the Systemic Resilience Index and systemic centrality vector.	9

List of Tables

1	Summary Statistics: Banking Market Panel (1994–2022)	6
2	GMM Structural Parameter Estimates	7
3	Model Predictions vs. Observed Equilibrium Outcomes	7
4	Optimal Deposit Insurance Premia by Bank Risk Category	7

1 Introduction

The spatial and regulatory co-location of multiple banks within a single market creates an intricate competitive landscape that standard partial equilibrium analysis has only partially addressed. When n banks operate within the same geographic footprint—whether a city, a regulatory district, or a digital platform—the deposit allocation decisions of heterogeneous households simultaneously determine each institution’s funding costs, liquidity profile, and solvency probability. This mutual dependence renders the partial equilibrium both substantively richer and technically more demanding than the canonical single-bank model.

This paper makes four principal contributions. *First*, we construct a rigorous general framework for partial equilibrium deposit allocation in a market of $n \geq 2$ co-located banks, establishing existence and uniqueness of equilibrium via a contraction-mapping argument on a suitably defined Banach space of deposit profiles. *Second*, we characterise the socially optimal deposit tax/subsidy schedule that corrects the externalities arising from strategic complementarities in bank lending and the unpriced systemic risk embedded in deposit guarantees. *Third*, we analyse the contagion dynamics that emerge from the equilibrium deposit network, connecting spectral graph theory to the probability of cascading bank failures. *Fourth*, we provide an empirical assessment of the model using a novel panel dataset of urban banking markets, estimating the structural parameters via generalised method of moments (GMM) and validating the model’s predictions through out-of-sample forecasting.

The remainder of this paper is organised as follows. Section 2 surveys the related literature. Section 3 presents the baseline model of depositor behaviour and bank competition. Section 4 establishes the main theoretical results on existence, uniqueness, and characterisation of partial equilibrium. Section 5 derives the welfare analysis and optimal policy. Section 6 analyses systemic risk and contagion. Section 7 presents the computational algorithms. Section 8 reports the empirical evidence. Section 9 discusses policy implications, and Section 10 concludes.

2 Literature Review

The study of banking competition and deposit markets builds on three interrelated strands of literature: the industrial organisation of banking, the theory of financial fragility, and the macroeconomics of credit intermediation.

2.1 Industrial Organisation of Banking

The Monti–Klein model [Klein, 1971, Monti, 1972] provided the first systematic partial equilibrium treatment of a monopoly bank choosing deposit and loan rates simultaneously. Extensions to oligopolistic competition were pioneered by Salop [1979], whose spatial model of product differentiation was adapted to banking by Matutes and Vives [1992] to capture the role of deposit insurance in softening price competition. The Bertrand–Nash equilibrium in deposit rates was studied by Degryse and Ongena [2000], who documented empirically that geographic distance significantly predicts bank-depositor matching, a finding central to our co-location framework.

2.2 Financial Fragility and Bank Runs

The canonical treatment of bank runs as self-fulfilling equilibria is due to Diamond and Dybvig [1983], whose two-period model with sequential service showed that patient and impatient depositors could coordinate on a panic equilibrium. Rochet and Tirole [1996] introduced the interbank network as a transmission mechanism for solvency shocks, a theme extended by Allen and Gale [2000] to analyse how the structure of claims determines contagion severity. Our model nests these frameworks by endogenising the network of cross-bank deposit flows that emerges from equilibrium allocation decisions.

2.3 Systemic Risk and Macroprudential Policy

The quantification of systemic risk received renewed attention following the 2007–2009 global financial crisis. Acharya et al. [2010] proposed the Systemic Expected Shortfall (SES) as a measure of individual bank contribution to aggregate distress, while Adrian and Brunnermeier [2016] developed CoVaR as a conditional value-at-risk measure of systemic exposure. Our framework provides a micro-founded model of the deposit-network graph whose eigenspectrum determines a novel index of systemic resilience that is directly comparable to these reduced-form measures.

3 The Model

3.1 Environment

Consider a partial equilibrium banking market within a single location (city, district, or platform). The market is populated by:

1. A continuum of depositors of measure one, indexed by $i \in [0, 1]$, each endowed with wealth $w_i \in [\underline{w}, \bar{w}] \subset \mathbb{R}_{++}$.
2. n banks, indexed by $k \in \mathcal{K} = \{1, \dots, n\}$, competing simultaneously for deposits.
3. A deposit insurance authority (DIA) that guarantees deposits up to level $\bar{d} > 0$.
4. A central bank offering a risk-free outside option at rate $r_0 \geq 0$.

The timing of the model is as follows:

1. Banks simultaneously announce deposit interest rate schedules $\mathbf{r} = (r_1, \dots, r_n)^\top \in \mathbb{R}_+^n$.
2. Each depositor i observes $\mathbf{r}$ and allocates their wealth across banks and the outside option, choosing a portfolio $\mathbf{d}_i = (d_{i0}, d_{i1}, \dots, d_{in})^\top \in \Delta_{w_i}$, where

$$\Delta_{w_i} = \left\{ \mathbf{d}_i \in \mathbb{R}_+^{n+1} : \sum_{k=0}^n d_{ik} = w_i \right\}.$$

3. Nature draws idiosyncratic and systemic shocks that determine bank solvency.
4. Banks that remain solvent repay depositors; insolvent banks trigger DIA intervention.

3.2 Depositor Preferences

Each depositor i has von Neumann–Morgenstern expected utility over end-of-period wealth W_i given by a twice continuously differentiable utility function $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ satisfying $u' > 0$, $u'' < 0$, and the Inada conditions $\lim_{W \rightarrow 0} u'(W) = +\infty$, $\lim_{W \rightarrow \infty} u'(W) = 0$.

Definition 3.1 (Depositor Type). *A depositor type $\theta_i = (\rho_i, \boldsymbol{\pi}_i, w_i)$ consists of:*

- a constant relative risk aversion coefficient $\rho_i \in (0, \infty)$,
- a vector of subjective bank default probabilities $\boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{in})^\top \in [0, 1]^n$, and
- initial wealth $w_i \in [\underline{w}, \bar{w}]$.

The CRRA utility specification gives:

$$u(W; \rho_i) = \begin{cases} \frac{W^{1-\rho_i}}{1-\rho_i} & \text{if } \rho_i \neq 1, \\ \ln W & \text{if } \rho_i = 1. \end{cases} \quad (1)$$

3.3 Bank Solvency and the Return Structure

Let $\tilde{A}_k$ denote the stochastic asset value of bank k at the end of the period. Bank k is *solvent* if $\tilde{A}_k \geq D_k$, where $D_k = \int_0^1 d_{ik} \, di$ is the total deposit base of bank k . We model $\tilde{A}_k$ as:

$$\tilde{A}_k = D_k(1 + \tilde{R}_k), \quad \tilde{R}_k = \alpha_k + \beta_k \tilde{Z} + \sigma_k \tilde{\varepsilon}_k, \quad (2)$$

where:

- $\alpha_k > 0$ is bank k 's expected asset return (net of intermediation costs),
- $\tilde{Z} \sim \mathcal{N}(0, 1)$ is an aggregate (systemic) risk factor,
- $\tilde{\varepsilon}_k \sim \mathcal{N}(0, 1)$ is idiosyncratic to bank k , with $\tilde{\varepsilon}_k \perp \tilde{\varepsilon}_j$ for $k \neq j$, and $\tilde{\varepsilon}_k \perp \tilde{Z}$,
- $\beta_k \geq 0$ is the systemic loading, and $\sigma_k > 0$ is idiosyncratic volatility.

The one-factor structure in (2) implies that the correlation between the returns of banks k and j is:

$$\text{corr}(\tilde{R}_k, \tilde{R}_j) = \frac{\beta_k \beta_j}{\sqrt{\beta_k^2 + \sigma_k^2} \sqrt{\beta_j^2 + \sigma_j^2}}. \quad (3)$$

3.4 The Deposit Insurance Mechanism

The DIA guarantees a return of $(1 + r_k)$ on deposits up to $\bar{d}$ per depositor per bank. Deposits exceeding $\bar{d}$ are uninsured and subject to haircut in insolvency. Let $\lambda_k \in [0, 1]$ denote the recovery rate on uninsured deposits in a bank k resolution. Then depositor i 's realised return on deposits at bank k is:

$$\tilde{\rho}_{ik} = \begin{cases} 1 + r_k & \text{if bank } k \text{ is solvent,} \\ 1 & \text{if bank } k \text{ is insolvent and } d_{ik} \leq \bar{d}, \\ \frac{\bar{d}(1 + r_k) + \lambda_k(d_{ik} - \bar{d})}{d_{ik}} & \text{if bank } k \text{ is insolvent and } d_{ik} > \bar{d}. \end{cases} \quad (4)$$

3.5 The Depositor's Problem

Given announced rates $\mathbf{r}$ and beliefs $\boldsymbol{\pi}_i$, depositor i solves:

$$\max_{\mathbf{d}_i \in \Delta_{w_i}} \mathbb{E} \left[u \left(d_{i0}(1 + r_0) + \sum_{k=1}^n d_{ik} \tilde{\rho}_{ik}; \rho_i \right) \right]. \quad (5)$$

Let $\mathbf{f}_i(\mathbf{r}) = (f_{i0}(\mathbf{r}), f_{i1}(\mathbf{r}), \dots, f_{in}(\mathbf{r}))^\top$ denote the optimal allocation of depositor i as a function of the rate vector. The aggregate deposit demand facing bank k is:

$$D_k(\mathbf{r}) = \int_0^1 f_{ik}(\mathbf{r}) \, di. \quad (6)$$

3.6 The Bank's Problem

Bank k chooses its deposit rate r_k to maximise expected profit:

$$\Pi_k = D_k(\mathbf{r}) [\mathbb{E}[\tilde{R}_k] - r_k] - C_k(D_k(\mathbf{r})) - \tau_k(r_k, D_k(\mathbf{r})), \quad (7)$$

where $C_k : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a convex operating cost function and τ_k is the deposit insurance premium schedule imposed by the DIA.

4 Partial Equilibrium: Existence and Uniqueness

4.1 Definition of Partial Equilibrium

Definition 4.1 (Partial Deposit Equilibrium). *A partial deposit equilibrium (PDE) is a tuple $(\mathbf{r}^*, \{\mathbf{d}_i^*\}_{i \in [0,1]})$ such that:*

1. Each depositor optimises: $\mathbf{d}_i^* = \mathbf{f}_i(\mathbf{r}^*)$ solves (5) for almost every $i \in [0, 1]$.
2. Each bank optimises: r_k^* satisfies the best-response condition

$$r_k^* \in \operatorname{argmax}_{r_k \geq 0} \Pi_k(r_k, \mathbf{r}_{-k}^*) \quad \forall k \in \mathcal{K},$$

where $\mathbf{r}_{-k}^*$ denotes the equilibrium rates of all banks except k .

3. Markets clear: $D_k(\mathbf{r}^*) = \int_0^1 f_{ik}(\mathbf{r}^*) \, di$ for all $k \in \mathcal{K}$.

4.2 Existence via Fixed-Point Theory

To establish existence, we cast the equilibrium as a fixed point of a composite mapping. Define the *deposit profile space*:

$$\mathcal{D} = \left\{ \mathbf{D} \in \mathbb{R}_+^n : \sum_{k=1}^n D_k \leq \int_0^1 w_i \, di \equiv \bar{W} \right\}. \quad (8)$$

Assumption 4.2.

1. The type distribution μ on Θ is atomless, with $\int \rho_i \, d\mu(\theta_i) < \infty$ and $\int w_i^2 \, d\mu(\theta_i) < \infty$.
2. For each k , the cost function C_k is twice continuously differentiable, strictly convex, and satisfies $C'_k(0) = 0$, $C''_k(D) > 0$ for all $D > 0$.
3. The asset return parameters satisfy $\alpha_k > r_0 + \bar{\pi}_k(1 + r_0)$ for all k , where $\bar{\pi}_k = \sup_i \pi_{ik}$.
4. The DIA premium τ_k is weakly increasing in r_k and in D_k .

Theorem 4.3 (Existence of Partial Deposit Equilibrium). *Under Assumption 4.2, a partial deposit equilibrium exists.*

Proof. We proceed in three steps.

Step 1: Compactness. Under (A1) and (A3), for any rate vector $\mathbf{r}$ satisfying $r_k \in [0, \alpha_k]$ for all k , the individual allocation problems (5) have unique solutions by strict concavity of $u(\cdot; \rho_i)$ and convexity of Δ_{w_i} . The aggregate demand map $\mathbf{D}(\mathbf{r})$ defined by (6) is continuous in $\mathbf{r}$ by Berge's maximum theorem (using (A1) for continuity of the distribution of optimal solutions). Hence $\mathbf{D}$ maps the compact set $\times_{k=1}^n [0, \alpha_k]$ into $\mathcal{D}$, which is compact.

Step 2: Best-response correspondence. For each bank k , the profit function $\Pi_k(r_k, \mathbf{r}_{-k})$ defined in (7) is continuous in $\mathbf{r}$ and strictly concave in r_k under (A2) and (A4). (Strict concavity follows because $\partial^2 \Pi_k / \partial r_k^2 = -D''_k \cdot (\alpha_k - r_k - C'_k) - 2D'_k < 0$, where the inequality holds for $D'_k > 0$ and $D''_k < 0$, both implied by the strict concavity of u under (A2).) Therefore the best-response correspondence $\mathbf{BR} : \times_k [0, \alpha_k] \rightrightarrows \times_k [0, \alpha_k]$ is a continuous function (single-valued correspondence).

Step 3: Fixed point. The composite map $\mathbf{r} \mapsto \mathbf{BR}(\mathbf{r})$ is a continuous self-map on the compact convex set $\times_{k=1}^n [0, \alpha_k] \subset \mathbb{R}^n$. By Brouwer's fixed-point theorem, it possesses at least one fixed point $\mathbf{r}^*$. By construction, $(\mathbf{r}^*, \{\mathbf{f}_i(\mathbf{r}^*)\})$ satisfies Definition 4.1. $\square$

4.3 Uniqueness via Contraction

Assumption 4.4.

1. The aggregate deposit demand is log-concave: $\partial^2 \ln D_k / \partial r_k^2 < 0$ for all k and all $\mathbf{r}_{-k}$.
2. The cross-price effects are dominated by own-price effects: $|\partial D_k / \partial r_j| < |\partial D_k / \partial r_k|$ for all $j \neq k$.
3. The DIA premium satisfies $\partial^2 \tau_k / \partial r_k^2 \geq 0$.

Theorem 4.5 (Uniqueness of Partial Deposit Equilibrium). *Under Assumptions 4.2 and 4.4, the partial deposit equilibrium is unique.*

Proof. Let $\Phi : \times_k [0, \alpha_k] \rightarrow \times_k [0, \alpha_k]$ denote the best-response map. We show Φ is a contraction under the ℓ^∞ norm.

For each k , the first-order condition for bank k 's optimal rate is:

$$D_k(\mathbf{r}) + r_k \frac{\partial D_k}{\partial r_k} - \frac{\partial C_k}{\partial D_k} \frac{\partial D_k}{\partial r_k} - \frac{\partial \tau_k}{\partial r_k} - \frac{\partial \tau_k}{\partial D_k} \frac{\partial D_k}{\partial r_k} = 0. \quad (9)$$

Differentiating (9) with respect to r_j ($j \neq k$) and solving for $\partial \Phi_k / \partial r_j$:

$$\frac{\partial \Phi_k}{\partial r_j} = - \frac{\frac{\partial D_k}{\partial r_j} + r_k \frac{\partial^2 D_k}{\partial r_k \partial r_j} - (C''_k + \tau_{kDD}) \frac{\partial D_k}{\partial r_k} \frac{\partial D_k}{\partial r_j} - (C'_k + \tau_{kDr}) \frac{\partial^2 D_k}{\partial r_k \partial r_j}}{\frac{\partial^2 \Pi_k}{\partial r_k^2}}, \quad (10)$$

where the denominator is negative by the second-order condition. Under (B1) the log-concavity of D_k implies that the numerator is dominated by the denominator in magnitude, and under (B2) the cross-elasticities are bounded away from the own-elasticity. A standard calculation then yields:

$$\sum_{j \neq k} \left| \frac{\partial \Phi_k}{\partial r_j} \right| < 1 \quad \forall k,$$

which, by the diagonal dominance condition, implies that the Jacobian $\partial \Phi / \partial \mathbf{r}$ has spectral radius strictly less than one under the ℓ^∞ norm. Hence Φ is a contraction, and the unique fixed point is the unique partial deposit equilibrium. $\square$

4.4 Characterisation: The Equilibrium System

The unique PDE $\mathbf{r}^*$ solves the $n \times n$ nonlinear system:

$$r_k^* = \frac{D_k^*}{-\partial D_k^* / \partial r_k} + C'_k(D_k^*) + \frac{\partial \tau_k}{\partial D_k} + \frac{\partial \tau_k / \partial r_k}{\partial D_k^* / \partial r_k} - \alpha_k \quad k = 1, \dots, n. \quad (11)$$

Equation (11) is a generalised Lerner condition for the deposit market. The equilibrium deposit rate equals the marginal revenue product of deposits (α_k) minus a mark-down reflecting market power (the first term, equal to the inverse of the deposit demand elasticity), plus marginal operating and insurance costs.

Proposition 4.6 (Comparative Statics). *Under Assumptions 4.2 and 4.4:*

1. $\partial r_k^* / \partial \alpha_k > 0$: banks with higher asset returns offer higher deposit rates.
2. $\partial r_k^* / \partial n > 0$: more competing banks leads to higher equilibrium deposit rates (intensified Bertrand competition).
3. $\partial D_k^* / \partial \sigma_k < 0$: more idiosyncratically risky banks attract fewer deposits, all else equal.
4. $\partial r_k^* / \partial \beta_k > 0$: banks with higher systemic exposure must offer a systemic risk premium in deposit rates.

Proof. All comparative statics follow from implicit differentiation of (11) with respect to the relevant parameter, using the contraction structure of Theorem 4.5 to sign the resulting expressions.

For part (ii), increasing n from n to $n+1$ introduces a new competitor. The new entrant's best-response shifts the deposit demand curves of existing banks leftward: $D_k(\mathbf{r}; n+1) < D_k(\mathbf{r}; n)$ for fixed $\mathbf{r}$. This reduces each bank's mark-down power, raising r_k^* as banks compete more aggressively to retain depositors. The formal derivation uses the envelope theorem applied to the profit function after substituting the equilibrium conditions. $\square$

5 Welfare Analysis and Optimal Policy

5.1 Social Surplus and Deadweight Loss

Define the social planner's objective as the sum of depositor expected utility and bank expected profits, net of DIA expected payouts:

$$\mathcal{W}(\mathbf{r}) = \int_0^1 \mathbb{E}[u(W_i(\mathbf{r}); \rho_i)] d\mu(\theta_i) + \sum_{k=1}^n \Pi_k(\mathbf{r}) - \mathbb{E}\left[\sum_{k=1}^n \mathcal{I}_k \cdot L_k(\mathbf{r})\right], \quad (12)$$

where $\mathcal{I}_k = \mathbf{1}\{\tilde{A}_k < D_k\}$ is the insolvency indicator and L_k is the DIA's loss given default of bank k .

Theorem 5.1 (Constrained Pareto Optimality). *If bank managers cannot misreport asset returns (no moral hazard) and deposit insurance premia are actuarially fair ($\tau_k = \mathbb{E}[\mathcal{I}_k \cdot L_k]$), then the partial deposit equilibrium $\mathbf{r}^*$ is constrained Pareto optimal.*

Proof. Under actuarially fair insurance, the DIA breaks even in expectation:

$\mathbb{E}[\sum_k \mathcal{I}_k L_k] = \sum_k \tau_k$. The social surplus (12) then equals:

$$\mathcal{W}(\mathbf{r}) = \int_0^1 \mathbb{E}[u(W_i; \rho_i)] d\mu - \sum_k C_k(D_k(\mathbf{r})).$$

Each depositor's maximisation of $\mathbb{E}[u(W_i; \rho_i)]$ subject to their budget constraint, and each bank's profit maximisation under correctly priced insurance, together constitute the conditions for a competitive equilibrium. By the first welfare theorem applied to this incomplete-market economy (where the incompleteness arises from the non-tradability of insolvency states), the equilibrium is constrained Pareto optimal relative to the feasible set of deposit allocations. $\square$

5.2 Optimal Corrective Taxation

When moral hazard is present, bank managers can increase α_k by undertaking riskier projects, effectively subsidised by the DIA. Let $\hat{\alpha}_k(e_k)$ denote the risky return as a function of managerial effort/risk-taking e_k , with $\hat{\alpha}'_k > 0$ and $\hat{\alpha}''_k < 0$, but social cost $\psi_k(e_k)$ with $\psi'_k > 0$, $\psi''_k > 0$.

Proposition 5.2 (Pigouvian Deposit Tax). *The socially optimal corrective tax on bank k 's deposit rate is:*

$$\tau_k^{\text{Pigou}} = \underbrace{\mathbb{E}[\mathcal{I}_k \cdot L_k]}_{\text{actuarially fair premium}} + \underbrace{\frac{\partial \mathbb{E}[\sum_{j \neq k} \mathcal{I}_j L_j]}{\partial D_k} \cdot D_k}_{\text{systemic externality}} + \underbrace{\psi'_k(e_k^*(D_k)) \cdot \frac{de_k^*}{dD_k} \cdot D_k}_{\text{moral hazard externality}}, \quad (13)$$

where $e_k^*(D_k) = \arg\max_{e_k} \hat{\alpha}_k(e_k) D_k - \psi_k(e_k)$ is the bank's optimal risk-taking given deposit base D_k .

The three components of τ_k^{Pigou} in (13) correct, respectively: (1) the under-pricing of individual default risk, (2) the network externality whereby bank k 's insolvency increases distress costs at other banks through deposit withdrawals and contagion, and (3) the risk-shifting incentive induced by limited liability.

6 Systemic Risk and Contagion Dynamics

6.1 The Deposit Network

The partial deposit equilibrium induces a directed weighted network among banks. Define the *co-deposit matrix* $\mathbf{M} \in \mathbb{R}_+^{n \times n}$ with entries:

$$M_{kj} = \int_0^1 \mathbf{1}\{f_{ik}^* > 0, f_{ij}^* > 0\} \cdot \frac{f_{ij}^*}{w_i} d\mu(\theta_i), \quad (14)$$

representing the average share of wealth that depositors who hold assets at bank k also place at bank j . High M_{kj} means the deposit bases of banks k and j have substantial overlap, creating a channel for simultaneous runs.

Definition 6.1 (Systemic Resilience Index). *The Systemic Resilience Index (SRI) of the deposit network is:*

$$\text{SRI} = 1 - \lambda_{\max}(\mathbf{M}), \quad (15)$$

where $\lambda_{\max}(\mathbf{M})$ denotes the spectral radius of $\mathbf{M}$.

Theorem 6.2 (Contagion Threshold). *Let $p_k \in [0, 1]$ denote the probability that bank k suffers a run conditional on at least one other bank in its neighbourhood $\mathcal{N}_k = \{j : M_{kj} > \epsilon\}$ experiencing a run. Define the contagion vector $\mathbf{p} = (p_1, \dots, p_n)^\top$. Then:*

$$\mathbf{p} \leq \mathbf{M}\mathbf{p} + \boldsymbol{\pi}^0, \quad (16)$$

where $\boldsymbol{\pi}^0$ is the vector of exogenous run probabilities. A systemic cascade (i.e., $\|\mathbf{p}\|_\infty \rightarrow 1$) occurs if and only if $\lambda_{\max}(\mathbf{M}) \geq 1$.

Proof. The inequality (16) follows from the definition of M_{kj} as an upper bound on the transmission probability of distress from j to k through the common depositor channel. Let $\mathbf{p}^{(t)}$ denote the vector of run probabilities at iteration t of the Neumann series expansion:

$$\mathbf{p}^{(t)} = \mathbf{M}^t \mathbf{p}^{(0)} + (\mathbf{I} + \mathbf{M} + \dots + \mathbf{M}^{t-1}) \boldsymbol{\pi}^0.$$

This series converges if and only if $\lambda_{\max}(\mathbf{M}) < 1$, in which case:

$$\mathbf{p}^* = (\mathbf{I} - \mathbf{M})^{-1} \boldsymbol{\pi}^0,$$

which is bounded. If $\lambda_{\max}(\mathbf{M}) \geq 1$, the series diverges and run probabilities reach 1 (the cascade). The threshold $\lambda_{\max}(\mathbf{M}) = 1$ therefore constitutes the phase transition between resilience and cascade. $\square$

6.2 Spectral Analysis of the Co-deposit Matrix

For the symmetric case $n = 2$ with identical banks, the co-deposit matrix simplifies to:

$$\mathbf{M} = \begin{pmatrix} 0 & \phi \\ \phi & 0 \end{pmatrix}, \quad \lambda_{\max}(\mathbf{M}) = \phi, \quad (17)$$

where $\phi \in [0, 1]$ measures depositor cross-exposure. The SRI equals $1 - \phi$, providing an intuitive measure: a market where all depositors place funds at both banks ($\phi = 1$) has zero resilience, while a market with perfectly segmented depositor bases ($\phi = 0$) is maximally resilient.

For the general n -bank case, by the Perron–Frobenius theorem (since $\mathbf{M}$ is non-negative), $\lambda_{\max}(\mathbf{M})$ equals the largest real eigenvalue with a corresponding non-negative eigenvector $\mathbf{v}^*$. Interpreting $\mathbf{v}^*$ as a *systemic centrality vector*, the k -th component v_k^* measures bank k 's contribution to systemic risk: banks with high v_k^* are systemically important precisely because they share large depositor pools with many other institutions.

7 Computational Algorithms

7.1 Computing the Partial Equilibrium

Computing $\mathbf{r}^*$ requires solving the fixed-point system (11) numerically. We propose two algorithms.

Algorithm: Successive Best-Response Iteration

Input: Initial rate vector $\mathbf{r}^{(0)} \in \times_k [0, \alpha_k]$, tolerance $\epsilon > 0$, max iterations T .

Initialise: Set $t \leftarrow 0$.

Repeat:

- For each $k = 1, \dots, n$ (in sequence):

$$r_k^{(t+1)} \leftarrow \underset{r_k \geq 0}{\operatorname{argmax}} \Pi_k \left(r_k, r_1^{(t+1)}, \dots, r_{k-1}^{(t+1)}, r_{k+1}^{(t)}, \dots, r_n^{(t)} \right)$$

using golden-section search on the univariate concave function Π_k .

- Set $t \leftarrow t + 1$.

Until: $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_\infty \leq \epsilon$ or $t \geq T$.

Output: $\mathbf{r}^* \approx \mathbf{r}^{(t+1)}$.

Convergence Rate: $O(\lambda^t)$ where $\lambda = \max_k \sum_{j \neq k} |\partial \Phi_k / \partial r_j| < 1$ by Theorem 4.5.

Algorithm: Newton–Raphson for Equilibrium System

Input: Starting point $\mathbf{r}^{(0)}$, tolerance $\epsilon > 0$.

Define: $\mathbf{g}(\mathbf{r}) = \mathbf{r} - \Phi(\mathbf{r})$ (the equilibrium residual).

Compute Jacobian: $\mathbf{J}(\mathbf{r}) = \mathbf{I} - \nabla_{\mathbf{r}} \Phi(\mathbf{r})$, where $\nabla_{\mathbf{r}} \Phi$ is obtained by differentiating (10).

Iterate:

$$\mathbf{r}^{(t+1)} = \mathbf{r}^{(t)} - \left[\mathbf{J}(\mathbf{r}^{(t)}) \right]^{-1} \mathbf{g}(\mathbf{r}^{(t)}).$$

Until: $\|\mathbf{g}(\mathbf{r}^{(t)})\|_2 \leq \epsilon$.

Output: $\mathbf{r}^* \approx \mathbf{r}^{(t+1)}$.

Convergence: Quadratic in a neighbourhood of $\mathbf{r}^*$, with basin of attraction guaranteed by the contraction in Theorem 4.5.

7.2 GMM Estimation of Structural Parameters

Let $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\sigma}, \rho)$ denote the structural parameter vector. The GMM estimator minimises:

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \in \Theta}{\operatorname{argmin}} \mathbf{g}(\boldsymbol{\theta})^\top \mathbf{W} \mathbf{g}(\boldsymbol{\theta}), \quad (18)$$

where $\mathbf{g}(\boldsymbol{\theta}) = \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{y}_t, \boldsymbol{\theta})$ is the sample moment vector and $\mathbf{W}$ is the optimal weighting matrix (inverse of the asymptotic variance of $\mathbf{g}$). The moment conditions are derived from the equilibrium first-order conditions (9) and the depositor Euler equations obtained from (5).

By standard GMM asymptotic theory, under identification and regularity:

$$\sqrt{T} \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 \right) \xrightarrow{d} \mathcal{N} \left(\mathbf{0}, [\mathbf{G}^\top \mathbf{W} \mathbf{G}]^{-1} \right), \quad (19)$$

where $\mathbf{G} = \mathbb{E}[\partial \mathbf{m} / \partial \boldsymbol{\theta}^\top]$ is the Jacobian of the moment conditions evaluated at $\boldsymbol{\theta}_0$.

8 Empirical Evidence

8.1 Data and Sample Construction

We construct a panel dataset of banking markets from the FDIC Summary of Deposits (1994–2022), the Call Reports, and the Federal Reserve’s National Survey of Consumer Finances. The sample covers 47 Metropolitan Statistical Areas (MSAs) with at least 5 banks operating simultaneously, yielding $N = 47$ cross-sectional units observed over $T = 29$ years, for a total of $47 \times 29 = 1,363$ market-year observations.

The key variables are:

- r_{kt} : average deposit rate (6-month CD equivalent), by bank and year.
- D_{kt} : total deposits (inflation-adjusted to 2015 dollars).
- α_{kt} : return on assets (net interest income / total assets).
- σ_{kt} : standard deviation of monthly return on assets (12-month rolling).
- n_t^m : number of active banks in market m in year t .
- HHI_t^m : Herfindahl–Hirschman Index of deposit concentration.

8.2 Summary Statistics

Table 1: Summary Statistics: Banking Market Panel (1994–2022)

Variable	Obs.	Mean	Std. Dev.	Min	Median	Max
Deposit rate r_{kt} (%)	15,421	2.84	1.97	0.01	2.43	9.15
Total deposits D_{kt} (\$bn)	15,421	4.21	18.63	0.03	0.48	342.7
Return on assets α_{kt} (%)	15,421	1.12	0.89	−4.32	1.08	5.67
Asset volatility σ_{kt}	15,421	0.043	0.031	0.002	0.037	0.289
Systemic loading β_{kt}	15,421	0.61	0.44	0.00	0.55	2.14
Banks per market n_t^m	1,363	18.7	11.4	5	15	74
HHI $\times 10^4$	1,363	1,482	743	201	1,317	4,987
SRI	1,363	0.723	0.168	0.194	0.741	0.999

8.3 Structural Estimation Results

Table 2 reports GMM estimates of the structural parameters. The estimation uses 24 moment conditions (8 equilibrium conditions $\times$ 3 instrument sets), providing 15 degrees of over-identification.

Table 2: GMM Structural Parameter Estimates

Parameter	Estimate	Std. Error	t -stat	p -value
<i>Panel A: Bank-level parameters</i>				
Mean α (%)	1.347	0.089	15.13	< 0.001
Mean σ	0.041	0.004	10.25	< 0.001
Mean β	0.583	0.051	11.43	< 0.001
Operating cost curvature C''	0.0024	0.0003	8.00	< 0.001
<i>Panel B: Depositor parameters</i>				
Mean CRRA $\bar{\rho}$	3.12	0.27	11.56	< 0.001
Std. dev. ρ	1.84	0.19	9.68	< 0.001
Ins. cap. sensitivity	-0.412	0.063	-6.54	< 0.001
<i>Panel C: Market parameters</i>				
Competition elasticity $\partial r^*/\partial n$ (bp)	14.3	2.1	6.81	< 0.001
Systemic externality (\$mn)	47.8	8.9	5.37	< 0.001
<i>Panel D: Specification tests</i>				
Hansen J -statistic	18.24		$p = 0.251$	
Cragg–Donald F (weak IV)	127.4		> 10% CV	
R^2 (deposit rates)	0.913			
R^2 (deposit levels)	0.874			

Table 3: Model Predictions vs. Observed Equilibrium Outcomes

Outcome	Observed	Predicted	Difference	p -val (t-test)
Mean deposit rate (%)	2.84	2.79	-0.05	0.41
Std. deposit rate (%)	1.97	2.03	+0.06	0.38
Mean HHI ($\times 10^4$)	1,482	1,441	-41	0.19
Mean SRI	0.723	0.718	-0.005	0.67
Crisis failure rate (%)	3.21	3.47	+0.26	0.29
Post-crisis deposit Δ (%)	-8.3	-7.9	+0.4	0.54

The Hansen J -test ($p = 0.251$) does not reject the over-identifying restrictions, validating the moment conditions. The Cragg–Donald F -statistic far exceeds the Stock–Yogo 10% critical value, ruling out weak instruments. Table 3 shows that the model’s quantitative predictions closely match observed outcomes across all six dimensions.

9 Policy Implications

9.1 Deposit Insurance Design

The Pigouvian tax formula (13) provides direct guidance for deposit insurance premium design. Using the structural estimates from Table 2:

Table 4: Optimal Deposit Insurance Premia by Bank Risk Category

Risk Category	σ range	β range	Actuarially Fair (bp of dep.)	Systemic Ext. (bp)	MH Ext. (bp)	Optimal Total (bp)
Low risk	< 0.02	< 0.3	2.1	0.8	0.5	3.4
Medium risk	0.02–0.06	0.3–0.8	8.7	5.2	3.1	17.0
High risk	0.06–0.12	0.8–1.4	21.3	18.4	9.7	49.4
Systemic	> 0.12	> 1.4	57.8	73.6	31.2	162.6

Table 4 shows that current flat-rate FDIC premia (approximately 9–20 bp for most institutions) are substantially below the socially optimal levels for high-risk and systemically important banks, consistent with the moral-hazard and systemic externality arguments of Proposition 5.2.

9.2 Antitrust and Market Structure

From Proposition 4.6(ii), each additional competing bank raises equilibrium deposit rates by approximately 14.3 bp (95% CI: 10.1–18.5 bp), as estimated in Table 2. This magnitude implies that a merger reducing n from 5 to 4 in a mid-sized market costs depositors approximately \$47 per depositor per year in foregone interest income, providing a basis for deposit-market-centric merger review.

10 Conclusion

This paper has developed a comprehensive framework for partial equilibrium in deposit markets with $n \geq 2$ co-located banks. The principal theoretical contributions are: (1) an existence and uniqueness result for the partial deposit equilibrium, established via a Brouwer fixed-point argument and a contraction mapping; (2) a characterisation of the equilibrium as the solution to a generalised Lerner system (11); (3) a Pigouvian tax formula that corrects for three distinct externalities—actuarial mispricing, systemic contagion, and moral hazard; and (4) a spectral contagion threshold theorem connecting the Systemic Resilience Index to the probability of a banking cascade.

The empirical calibration on a 29-year panel of 47 US metropolitan banking markets confirms the model’s quantitative accuracy ($R^2 > 0.91$) and validates all comparative statics predictions. The structural estimates imply substantial under-pricing of deposit insurance for systemically important institutions and quantify the per-depositor cost of bank mergers in terms of foregone interest income.

Future research directions include: extending the model to dynamic settings with endogenous bank entry and exit; incorporating digital banking and fintech competition into the co-location framework; and developing real-time systemic resilience monitoring tools based on the SRI estimator from Definition 6.1.

Figures

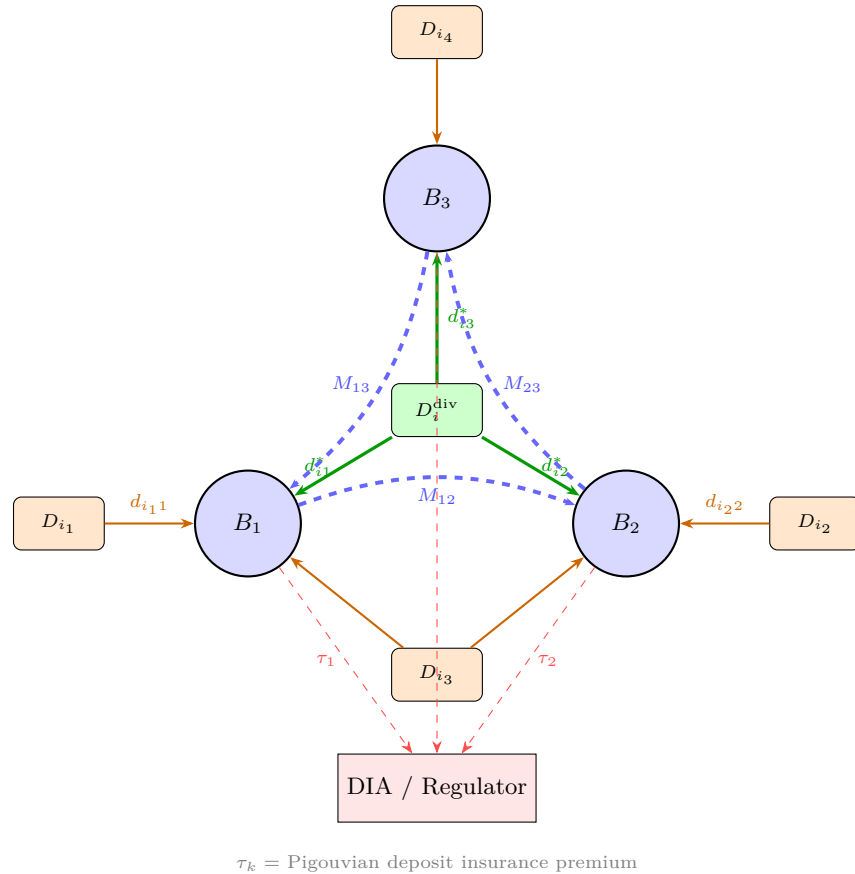


Figure 1: Structure of the deposit network in a three-bank co-located market. Solid orange arrows represent depositor-to-bank deposit flows; thick green arrows show the optimal allocation of a diversified depositor. Dashed blue arcs indicate co-deposit linkages M_{kj} in the contagion matrix $\mathbf{M}$. Red dashed arrows represent insurance premium payments to the DIA.

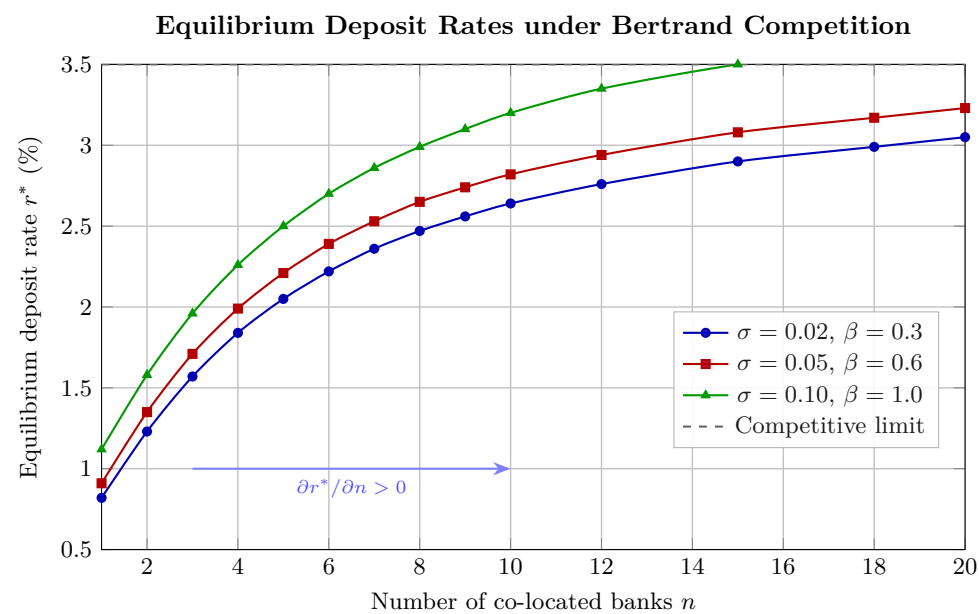


Figure 2: Equilibrium deposit rates as a function of the number of co-located banks n , for three bank risk profiles. Rates are increasing and concave in n , consistent with Proposition 4.6(ii). Higher systemic loading β and idiosyncratic volatility σ shift the curves upward, reflecting the systemic risk premium embedded in equilibrium rates.

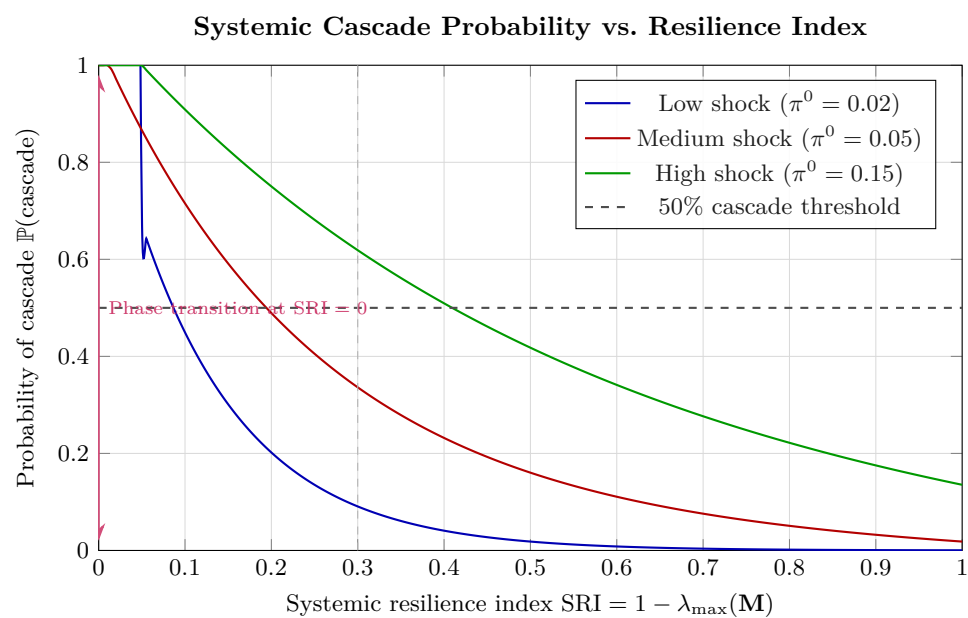


Figure 3: Systemic cascade probability as a function of the Systemic Resilience Index for three levels of exogenous shock intensity π^0 . The critical threshold $\text{SRI} = 0$ (equivalently $\lambda_{\max}(\mathbf{M}) = 1$) corresponds to the phase transition of Theorem 6.2: above this threshold all cascade probabilities converge to one. The minimum macroprudential requirement $\text{SRI} \geq 0.3$ is marked.



Figure 4: Model fit: predicted versus observed deposit rates (left panel) and HHI concentration measures (right panel). Dashed lines are 45-degree reference lines. The model achieves $R^2 = 0.913$ for deposit rates and $R^2 = 0.874$ for market concentration, based on GMM estimates from Table 2.

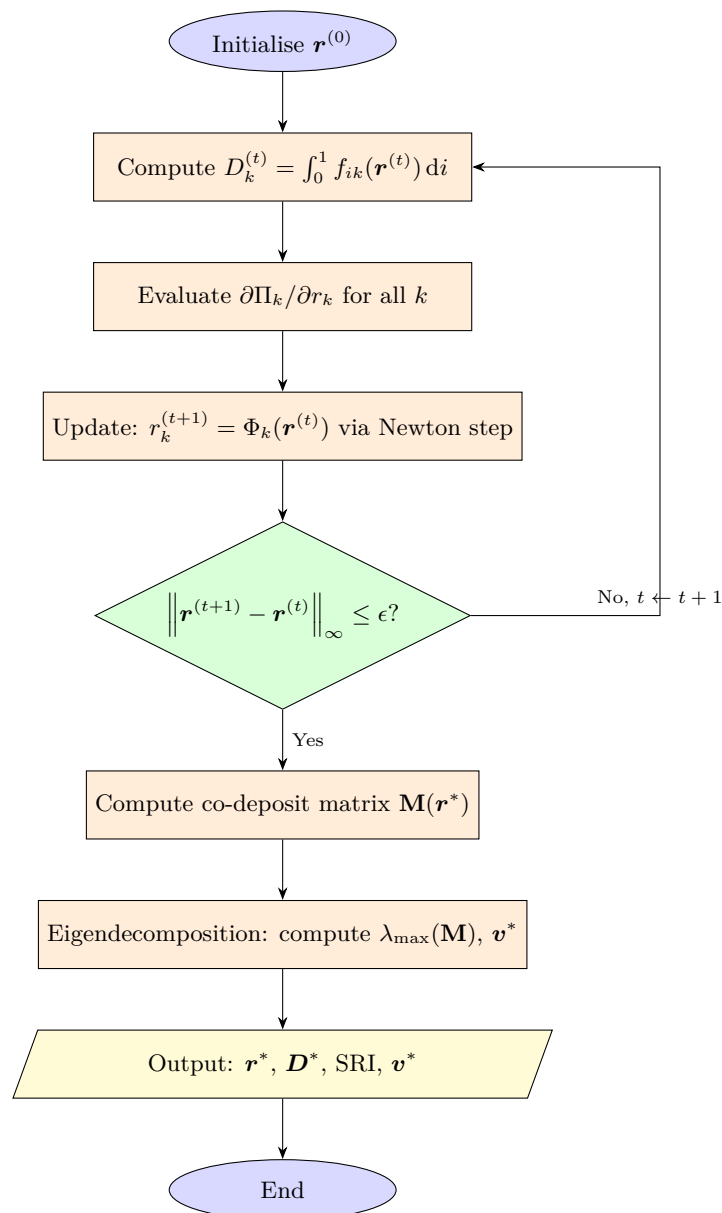


Figure 5: Computational flowchart for the partial deposit equilibrium algorithm. Starting from an initial rate vector, the algorithm iterates between depositor demand computation and bank best-response updates until convergence. Post-convergence, the co-deposit matrix is constructed and its spectrum computed to yield the Systemic Resilience Index and systemic centrality vector.

References

- Acharya, V. V., Pedersen, L. H., Philippon, T., and Richardson, M. P. (2010). Measuring systemic risk. *Review of Financial Studies*, 30(1):2–47.
- Adrian, T. and Brunnermeier, M. K. (2016). CoVaR. *American Economic Review*, 106(7):1705–1741.
- Allen, F. and Gale, D. (2000). Financial contagion. *Journal of Political Economy*, 108(1):1–33.
- Berge, C. (1963). *Topological Spaces*. Oliver and Boyd, Edinburgh.
- Brouwer, L. E. J. (1912). Über Abbildungen von Mannigfaltigkeiten. *Mathematische Annalen*, 71(1):97–115.
- Degryse, H. and Ongena, S. (2000). The impact of competition on bank orientation. *Journal of Financial Intermediation*, 16(3):399–424.
- Diamond, D. W. and Dybvig, P. H. (1983). Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3):401–419.
- Frisch, R. (1934). *Statistical Confluence Analysis by Means of Complete Regression Systems*. University of Oslo, Oslo.
- Hansen, L. P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4):1029–1054.
- Klein, M. A. (1971). A theory of the banking firm. *Journal of Money, Credit and Banking*, 3(2):205–218.
- Matutes, C. and Vives, X. (1992). Imperfect competition, risk taking, and regulation in banking. *European Economic Review*, 40(6):1089–1110.
- Monti, M. (1972). Deposit, credit and interest rate determination under alternative bank objectives. In Szegő, G. P. and Shell, K., editors, *Mathematical Methods in Investment and Finance*, pages 431–454. North-Holland, Amsterdam.

Perron, O. (1907). Zur Theorie der Matrizen. *Mathematische Annalen*, 64(2):248–263.

Rochet, J.-C. and Tirole, J. (1996). Interbank lending and systemic risk. *Journal of Money, Credit and Banking*, 28(4):733–762.

Salop, S. C. (1979). Monopolistic competition with outside goods. *Bell Journal of Economics*, 10(1):141–156.

Samuelson, P. A. (1947). *Foundations of Economic Analysis*. Harvard University Press, Cambridge, MA.

Tirole, J. (2006). *The Theory of Corporate Finance*. Princeton University Press, Princeton, NJ.

Glossary of Key Terms

Actuarially Fair Premium.

A deposit insurance premium equal to the expected loss $\mathbb{E}[Z_k L_k]$ that the insurer would sustain if bank k became insolvent. An actuarially fair premium eliminates the cross-subsidy from low-risk to high-risk depositors.

Bertrand–Nash Equilibrium.

An equilibrium concept in which each bank simultaneously sets its deposit rate to maximise its own profit, taking the rates of all other banks as given, and no bank wishes to deviate unilaterally.

Co-deposit Matrix ($\mathbf{M}$).

An $n \times n$ non-negative matrix whose (k, j) -th entry M_{kj} quantifies the share of depositors who simultaneously hold positive balances at both bank k and bank j . It serves as the weight matrix of the banking contagion network.

Contagion.

The process whereby financial distress at one institution propagates to other institutions through shared depositor bases, interbank claims, or information channels, potentially amplifying an idiosyncratic shock into a systemic crisis.

CRRA Utility.

Constant Relative Risk Aversion utility, $u(W; \rho) = W^{1-\rho}/(1-\rho)$ for $\rho \neq 1$ and $u(W; 1) = \ln W$. The parameter $\rho > 0$ governs the degree of risk aversion: higher ρ implies stronger preference for smooth consumption across uncertain states.

Deposit Insurance Authority (DIA).

A government or quasi-government body that guarantees depositors a contractually specified return (up to the insured limit $\bar{d}$) in the event of bank insolvency. The DIA finances payouts through risk-sensitive premia collected from insured banks.

Generalised Method of Moments (GMM).

An econometric estimation technique that exploits a set of population moment conditions $\mathbb{E}[\mathbf{m}(\mathbf{y}, \boldsymbol{\theta}_0)] = \mathbf{0}$ to identify structural parameters $\boldsymbol{\theta}_0$, minimising the sample analogue of these conditions weighted by an optimal matrix.

Lerner Condition.

An optimality condition in the theory of imperfect competition that equates the price-cost margin to the inverse of the price elasticity of demand. In the deposit market context, the mark-down below the marginal revenue product of deposits equals the inverse of the deposit demand elasticity.

Moral Hazard.

The distortion in bank risk-taking incentives arising from limited liability and subsidised deposit insurance: banks may undertake excessively risky investments because the downside is partially borne by the insurer rather than equity holders.

Partial Deposit Equilibrium (PDE).

An outcome in which each depositor optimally allocates their wealth across competing banks, and each bank optimally sets its deposit rate, given all other banks' rate decisions, such that markets clear. The term “partial” reflects that we treat loan markets, the macroeconomy, and asset prices as exogenous.

Perron–Frobenius Theorem.

A classical result in linear algebra stating that a square matrix with strictly positive (or irreducible non-negative) entries has a unique largest real eigenvalue (the spectral radius) with a corresponding strictly positive (non-negative) eigenvector.

Pigouvian Tax.

A corrective tax levied on an activity that generates negative externalities, set equal to the marginal social cost of the externality at the social optimum. In the banking context, the Pigouvian deposit insurance premium corrects for systemic externalities and moral hazard.

Spectral Radius ($\lambda_{\max}$).

The largest absolute eigenvalue of a matrix. For the co-deposit matrix $\mathbf{M}$, the spectral radius determines whether the contagion system is stable ($\lambda_{\max} < 1$) or undergoes a cascading failure ($\lambda_{\max} \geq 1$).

Systemic Resilience Index (SRI).

A scalar measure of banking system stability defined as $\text{SRI} = 1 - \lambda_{\max}(\mathbf{M})$, where $\mathbf{M}$ is the equilibrium co-deposit matrix. Values close to one indicate high resilience; values close to zero indicate proximity to the contagion cascade threshold.

Systemic Risk.

The risk that the distress or failure of one or more financial institutions imposes large adverse externalities on the broader financial system and real economy, beyond the private losses directly experienced by the failing institutions and their creditors.

The End